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Research paper

Prevalence of and factors associated with anxiety among school going adolescents analysis from 59 countries

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ABSTRACT

Background: Adolescence is a critical period marked by significant emotional and psychological changes, often accompanied by increased vulnerability to anxiety. This study aims to investigate the prevalence of anxiety and its potential factors among school-going adolescents across 59 countries.

Methods: We analyzed data from a comprehensive survey, the Global School-based Student Health Survey (GSHS), involving 179,937 adolescents aged 11—17 from 59 countries. Anxiety was measured based on self-reported experiences. Logistic regression analyses were employed to explore the association between anxiety while adjusting for different potential confounders.

Results: The prevalence of anxiety is greater in the eastern Mediterranean region, 14.61 % (95 % CI: 14.59 — 14.63), while the lowest prevalence is found in the Southeast Asian region, 3.78 % (95 % CI: 3.72 — 3.83). The second highest prevalence was found in the African region, which is almost four-fold higher than that of the lowest southeast Asian region, 11.43 % (95 % CI: 11.41 — 11.46). The logistic regression analysis reveals that age (especially older adolescents), gender (females), food insecurity, experiences of bullying, lack of close friends, parental understanding, sedentary behavior, and suicidal thoughts or actions are significant factors that contribute to anxiety among adolescents.

Conclusion: This study highlights by examining patterns across diverse cultural and economic settings, this research aims to inform targeted interventions and policy strategies that promote adolescent mental health worldwide.

1. Introduction

Adolescence is a pivotal developmental stage characterized by profound biological, psychological, and social transitions. During this period, mental health vulnerabilities—particularly anxiety disorders—often emerge, with long-lasting implications for academic performance, social relationships, and overall well-being. Addressing adolescents' psychological issues is paramount and essential for worldwide health and safety, with anxiety disorders being particularly significant. It encompasses a range of mental disorders, characterized by excessive fear, worry, obsessive thoughts, and sleep disturbances resulting from worry or fear (Khan and Khan, 2020). Anxiety disorders rank as the second foremost contributor to disability-adjusted life years (DALYs) and years lived with disability (YLDs) globally (Xiong et al., 2022). Adolescents aged 10–19, as determined by the United Nations, are especially susceptible to anxiety disorders because of the transitory phase between childhood and adulthood, which is characterized by physical, emotional, and psychosocial transformations (Liu et al., 2024). Anxiety disorders have escalated during the past three decades. Approximately 4.05 % of the global population, equivalent to 301 million individuals, suffers from anxiety. Over 55 % more individuals were impacted by anxiety from 1990 to 2019 (Javaid et al., 2023).

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Although, older adolescents exhibit a higher propensity for anxiety problems and but 4.4 % of individuals aged 10 to 14 and 5.5 % of those aged 15 to 19 experience anxiety disorders around the world (WHO, 2024). Globally, some regions expose a higher prevalence of anxiety disorders, with the highest rates in Latin America, the Caribbean, high-income regions of North America, and Western Europe. South Asia and sub-Saharan Africa exhibit the lowest prevalence of anxiety disorders (Javaid et al., 2023).

Adolescent anxiety symptoms are associated with impaired functioning, academic underperformance, emotional suffering, and diminished quality of life (Viswanathan et al., 2022). Furthermore, adolescents with anxiety problems frequently exhibit physiological, cognitive, and behavioral irregularities (Saiz Martínez et al., 2022). It increases the likelihood of diminished self-esteem, depression, tobacco consumption, substance dependence, and suicide. These impede adolescents' health and development (Lawrence et al., 2017). Prolonged anxiety can impact adolescents and hinder daily functioning, and adolescent anxiety disorders increase the probability of adult anxiety disorders (Kim and Kim, 2024). Parental education and school-based initiatives aimed at enhancing social and emotional learning, as well as coping mechanisms in children and adolescents, constitute excellent community-based strategies for anxiety prevention. Fitness regimens may also mitigate adult anxiety disorders (WHO, 2023). Anxiety interventions often encompass a blend of medication and psychotherapy (Sartori and Singewald, 2019).

Despite the growing recognition of adolescent anxiety as a global public health concern, cross-national data capturing both prevalence and associated risk factors remain limited. Factors such as gender, socioeconomic status, exposure to bullying, family structure, and parental mental health have been consistently linked to elevated anxiety risk. Moreover, disparities in mental health infrastructure and cultural perceptions of psychological distress contribute to underdiagnosis and undertreatment, particularly in low- and middle-income countries (LMICs). The study's large sample size, appropriate statistical modeling, global regions, and emphasis on a notably vulnerable region make it a significant contribution to global public health, social policy, and adolescent welfare and well-being. The results of this study may influence scholarly knowledge and practical solutions aimed at improving the lives of adolescents globally. The chief objective of this study is to thoroughly assess and quantify the prevalence of anxiety among 179,937 school-going adolescents across 59 countries, as well as to identify potential risk factors that may exacerbate these issues.

2. Methods

2.1. Data source and study design

This analysis made use of data collected from the Global School-based Student Health Survey (GSHS), which ran from 2003 to 2015. For this investigation, all publicly accessible, nationally representative datasets containing variables relevant to the analysis were chosen. The most recent GSHS dataset was chosen for nations having multiple datasets. Based on school and student response rates, county-level response rates were calculated. Children enrolled in school, aged 11 to 17, from both high-income counties (HIC) and low- and middle-income nations (LMICs) were among the participants. The GSHS used the same measures to collect data in each participating country, with certain modifications made to account for the socio-cultural circumstances unique to each nation (World Health Organization (WHO), 2021). Countries customized questionnaires by adding examples, choices, or phrases unique to their nation along with the basic core questions. The questionnaires were translated into the students' home tongues. To ensure understandability, the questionnaires were pilot-tested after being translated into the students' native tongue. The GSHS used a two-stage cluster sampling method to get a sample of students from all the involved countries that were representative of the whole group. In the beginning, schools were picked from a wide range of places within each country. A probability scale based on the number of students was used to pick a sample of these schools. Then, classes within the sample schools were chosen at random so that every student had an equal chance of being involved. Every student in those classes met the requirements. The WHO GSHS platform has a lot of information about survey tools, management, quality control, and sample designs (World Health Organization (WHO), 2021).

2.2. Ethics statement

The GSHS received ethical approval from the regulatory agencies of the participating country, including the Ministry of Education and other relevant institutional ethics review committees. Participation necessitates parental approval, either in writing or verbal form. The survey was administered anonymously.

2.3. Measures

2.3.1. Outcome variable: level of anxiety

Adolescents' level of anxiety was considered the main outcome variable for this study. Anxiety was assessed with, "*During the past 12 months, how often have you been so worried about something that you could not sleep at night*?" Response options were 'never', 'rarely', 'sometimes', 'most of the time' and 'always.' The answers 'most of the time' and 'always' were recoded as "yes," while 'never', 'rarely', and 'sometimes' were characterized as "no" to form binary variables.

2.3.2. Independent variables

2.3.2.1. Sociodemographic. The individuals' sex was classified as either male or female. Before it was reclassified, age was a continuous variable. The extent of food insecurity was measured by: "*During the past 30 days, how often did you go hungry because there was not enough food in your home*?' Responses of 'most of the time' or 'always' were recoded as having 'severe' food insecurity. Responses of 'rarely' or 'sometimes' as 'moderate' food insecurity, and 'never' as no food insecurity.

2.3.2.2. Peer victimization and conflict. To evaluate violence and other peer conflict, questions concerning the frequency and severity of physical injuries, involvement in physical altercations, and bullying were asked. "*Students were asked to report the number of times during the past 12 months they were physically attacked and/or in physical altercations*". All responses were categorized dichotomously, with individuals reporting multiple occurrences of any incident recoded as 'yes.'

2.3.2.3. Peer and parental support. Adolescent perceptions of peer support at school, parental understanding, parental monitoring, and parental control were indicators of social support. Peer supports were assessed by asking, "*During the past 30 days, how often were most of the students in your school kind and helpful?*" Three questions were used to assess parent-adolescent relationships: 1) "*During the past 30 days, how often did your parents or guardians understand your problems and worries*?" 2) "*During the past 30 days, how often did your parents or guardians really know what you were doing with your free time*?" 3) "*How often did your parents or guardians go through your things without your approval during the past 30 days*?" Responses for each item were 'never,' 'rarely,' 'sometimes,' 'most of the time,' and 'always.' These were recoded to trichotomous variables of "never/rarely," "sometimes," and "most of the time/always."

2.3.2.4. Health and lifestyle factors. Questions about sedentary habits, and obesity were included in the health and lifestyle questionnaire. Participants were asked about their weight and height, as well as the amount of time they spent doing stationary activities. Students were asked: "*How much time do you spend during a typical or usual day sitting and watching television, playing computer games, talking with friends, or doing other sedentary activities*." Answer choices were 'none,' '<1 h,' '1–2 h,' '3–4 h,' and '5 hours or more.' The height and weight information were self-reported instead of measured. The body mass index (BMI) for each participant was determined by dividing weight by the square of height (kg/m^2). Individual BMIs were thereafter compared to age- and sex-specific z-scores utilizing the 2007 WHO guidelines (World Health Organization (WHO), 2007). Study participants were categorized as possessing a normal BMI if their derived score was within two standard deviations below or one standard deviation above the WHO mean z-score. Overweight is defined as exceeding one standard deviation, whereas obesity is defined as exceeding two standard deviations. Loneliness was assessed with, "*How often have you felt lonely during the past 12 months?*" Suicidal ideation, planning, and attempt were already binary variables (e.g., "*Did you ever seriously consider attempting suicide during the past 12 months?*" with answers of "yes" or "no").

2.4. Statistical analysis

Weighted estimates of the mean number of psychosocial health problems were generated for national and regional data, along with the corresponding 95 % confidence intervals (CIs). Sampling weights were implemented to guarantee that the estimate was nationally representative. The analytical investigation employed a composite sample that accounted for the country-specific primary sampling unit (PSU), stratum, and sample weight, because of the data's complexity. All analyses were weighted using a PSU that was determined by the probability of selecting a school, a classroom, non-response at the school and student levels, grade, and gender. Binary logistic regression was applied to identify the factors associated with anxiety among adolescents. The logistic regression model was estimated using a sample of 101,826 observations. The log-likelihood of the null model (containing only the intercept) was −7,283,192, whereas the log-likelihood of the fitted model was substantially improved at −6,404,011, indicating a strong enhancement in model fit with the inclusion of predictor variables. Model selection criteria further support the adequacy of the fitted model. The Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) were both approximately 12.8 million. Given that both AIC and BIC penalize model complexity, and the model included 29 estimated parameters, these values suggest a reasonable balance between goodness-of-fit and parsimony. To maintain consistency and clarity in numerical presentation, percentages are reported to two significant figures throughout the manuscript. This level of precision was chosen to align with conventions in similar epidemiological studies and to preserve interpretive distinctions in subgroup comparisons where values are closely spaced. We acknowledge that this may introduce a degree of pseudoprecision; however, we believe it enhances transparency and comparability across results.

3. Results

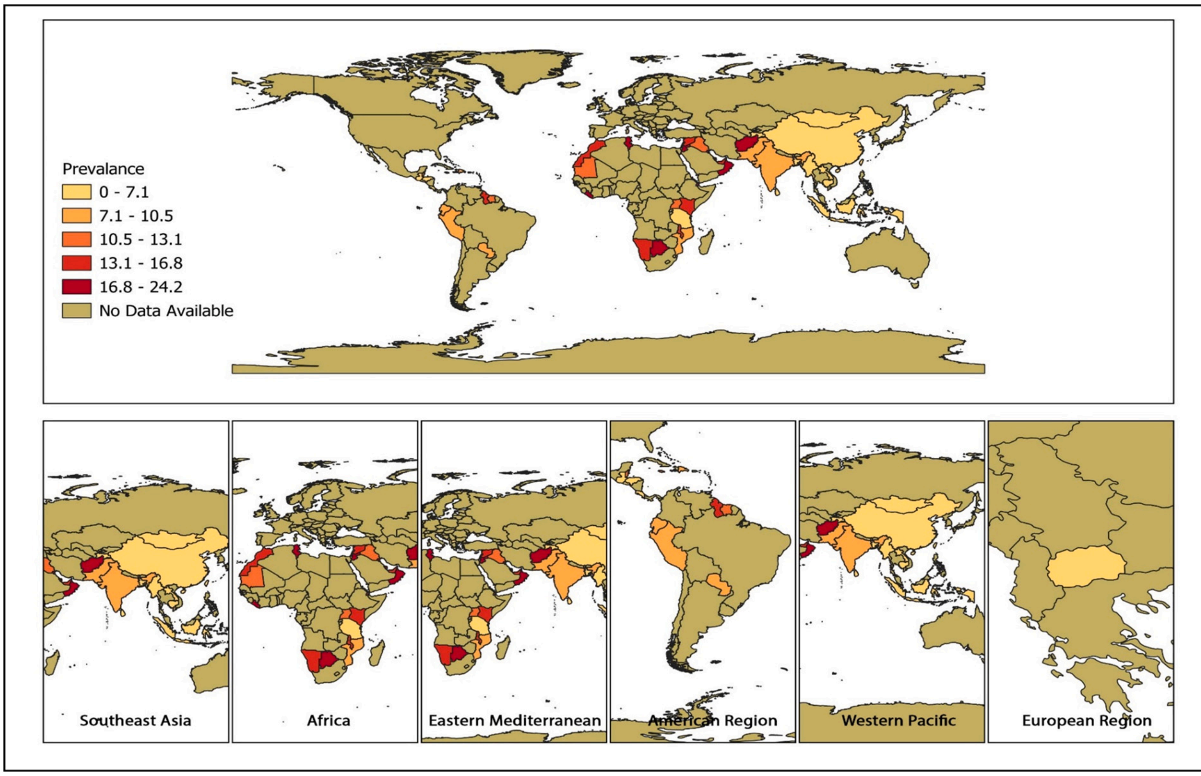
3.1. Country-wise participants status

Table 1 provides data on the prevalence of anxiety among adolescents in various countries, including the number of participants (N), the mean prevalence of anxiety, the standard deviation (St. Deviation), and the 95 % confidence interval (CI) for the mean prevalence. Data highlights significant variability in the prevalence of anxiety among adolescents across different countries. Countries like Afghanistan and Liberia show higher prevalence rates, while countries like Bangladesh and Indonesia show lower rates. The standard deviations and confidence intervals provide additional context on the variability and precision of these estimates. Afghanistan, Benin, Kuwait, Liberia and Tuvalu had the highest prevalence of men score while countries like Bahrain, Ghana, Jordan, Lebanon, Maldives, Morocco, Namibia, Saint Lucia, Trinidad and Tobago, Vanuatu, and Yemen fall into this category with mean prevalence ranging from 0.10 to 0.19. Countries like Anguilla, Bangladesh, Bhutan, Costa Rica, Honduras, Indonesia, Lao People's Dem, Myanmar, Nepal, Sri Lanka, Tanzania, and Uganda have lower mean prevalence, ranging from 0.04 to 0.10. (See Map 1.)

Table 2 highlights significant factors associated with adolescents' psychosocial health problems globally. As adolescents age, the prevalence of anxiety appears to increase. Among those aged 11–12, 6.84 % always report anxiety, which steadily rises to 23.87 % for those aged 17. Anxiety levels are significantly higher among females (60.04 %) compared to males (39.96 %). Adolescents who are bullied experience

Table 1
Participated country in the study with their mean and St. Deviation and corresponding 95 % CI.

	N	Mean	St. Deviation	95 % CI
Afghanistan	2514	0.25	0.43	(0.23—0.27)
Anguilla	797	0.10	0.30	(0.08—0.12)
Antigua and Barb	1252	0.14	0.35	(0.12—0.16)
Bahamas	1329	0.14	0.34	(0.12—0.16)
Bahrain	7113	0.15	0.35	(0.14—0.15)
Bangladesh	2978	0.05	0.21	(0.04—0.05)
Benin	2524	0.21	0.41	(0.19—0.22)
Bhutan	7565	0.08	0.28	(0.08—0.09)
British Virgin I	1653	0.11	0.31	(0.09—0.12)
Cook Islands	699	0.14	0.35	(0.12—0.17)
Costa Rica	2676	0.05	0.22	(0.04—0.06)
Curaçao	2729	0.11	0.32	(0.10—0.12)
Dominican Republic	1463	0.10	0.30	(0.09—0.12)
Ecuador	2206	0.08	0.27	(0.07—0.09)
El Salvador	1911	0.07	0.26	(0.06—0.08)
Fiji	3665	0.14	0.34	(0.13—0.15)
French Polynesia	3209	0.12	0.33	(0.11—0.13)
Ghana	5597	0.15	0.36	(0.14—0.16)
Honduras	1769	0.06	0.24	(0.05—0.07)
Indonesia	11,102	0.05	0.21	(0.04—0.05)
Iraq	2021	0.13	0.34	(0.12—0.15)
Jamaica	1646	0.13	0.34	(0.11—0.15)
Jordan	2143	0.18	0.38	(0.16—0.19)
Kenya	3547	0.14	0.35	(0.13—0.16)
Kiribati	1576	0.09	0.29	(0.08—0.11)
Kuwait	3497	0.21	0.41	(0.20—0.23)
Lao People's Dem	3669	0.05	0.21	(0.04—0.05)
Lebanon	5664	0.15	0.36	(0.14—0.16)
Liberia	2638	0.20	0.40	(0.19—0.22)
Maldives	3413	0.16	0.36	(0.14—0.17)
Mauritania	2025	0.12	0.32	(0.10—0.13)
Mauritius	2952	0.10	0.29	(0.09—0.11)
Mongolia	5369	0.06	0.23	(0.05—0.06)
Morocco	6660	0.18	0.38	(0.17—0.18)
Mozambique	1871	0.11	0.32	(0.10—0.13)
Myanmar	2815	0.04	0.19	(0.03—0.04)
Namibia	4486	0.15	0.36	(0.14—0.16)
Nepal	6488	0.04	0.20	(0.04—0.05)
Paraguay	3131	0.10	0.29	(0.08—0.11)
Peru	2864	0.09	0.29	(0.08—0.10)
Saint Kitts and	1725	0.09	0.29	(0.08—0.10)
Saint Lucia	1924	0.15	0.35	(0.13—0.16)
Samoa	1930	0.10	0.30	(0.09—0.11)
Seychelles	2503	0.11	0.32	(0.10—0.13)
Solomon Islands	1381	0.13	0.34	(0.11—0.15)
Sri Lanka	3225	0.05	0.21	(0.04—0.06)
St. Lucia	1273	0.12	0.32	(0.10—0.13)
Suriname	2115	0.12	0.33	(0.11—0.14)
Tanzania	3727	0.06	0.24	(0.05—0.07)
Timor-Leste	3669	0.12	0.32	(0.11—0.13)
Tokelau	140	0.11	0.32	(0.06—0.17)
Trinidad and Tob	3290	0.15	0.36	(0.14—0.16)
Tunisia	3849	0.14	0.35	(0.13—0.15)
Tuvalu	2811	0.20	0.40	(0.18—0.21)
Uganda	936	0.06	0.25	(0.05—0.08)
United Arab Emeritus	3202	0.11	0.32	(0.10—0.12)
Vanuatu	5763	0.17	0.37	(0.16—0.18)
Wallis and Futun	2147	0.07	0.25	(0.06—0.08)
Yemen	1101	0.15	0.36	(0.13—0.17)



Map 1. Prevalence of anxiety among school-going adolescents by different regions and countries

higher levels of anxiety, with 48.89 % always reporting it compared to 51.11 % who never do. Both physical attacks and physical fighting are linked to increased anxiety, with 43.90 % of physically attacked adolescents and 43.74 % of those involved in physical fighting always reporting anxiety. Serious injuries also show a stark difference, with 60.08 % of injured adolescents always feeling anxious compared to 39.92 % who never report it. A lack of peer support correlates with heightened anxiety. Adolescents who never receive peer support report anxiety 15.90 % of the time, while 44.91 % who receive support most of the time or always still experience anxiety. Lack of parental monitoring and understanding contributes to higher anxiety. Adolescents whose parents never check homework report 31.91 % always feeling anxious, and 34.02 % of those whose parents never understand their problems report anxiety. Adolescents with no friends are more likely to always feel anxious (11.29 %) than those with one or two (35.74 %) or more than three friends (52.97 %). Longer sitting times (>4 h per day) are associated with higher anxiety levels (28.29 %). In contrast, adolescents who spend less time sitting (<1 h) report anxiety less frequently (29.63 %). Adolescents who are overweight or obese report slightly higher anxiety (7.04 %) than those with normal weight (80.76 %). Adolescents with suicidal ideation (36.49 %), plans (31.94 %), or attempts (33.26 %) consistently report higher anxiety levels compared to those without these behaviors. All findings are statistically significant with *P*-values <0.001.

The (Table 3) shows adolescents aged 15, 16, and 17 are significantly more likely to experience psychosocial health problems compared to 11–12-year-olds, with odds increasing as age increases. For instance, 17-year-olds have 2.45 times higher psychosocial health issues compared to 11—12—year—olds (AOR: 2.45, 95 % CI: 2.05—2.94, $p < 0.001$). Female adolescents are more likely to experience psychosocial health problems than males (AOR: 1.51, 95 % CI: 1.38—1.66, $p < 0.001$). Adolescents who experience food insecurity “sometimes or rarely” or “most of the time or always” have higher odds of psychosocial health problems, with more severe food insecurity (AOR: 2.22, 95 % CI: 1.94—2.55, $p < 0.001$) showing a stronger association. Being bullied significantly increases the likelihood of psychosocial health problems (AOR: 1.68, 95 % CI: 1.52—1.85, $p < 0.001$) compared to those who were not bullied. Being involved in physical fights (AOR: 1.26, 95 % CI: 1.46—1.79, $p < 0.001$) and being seriously injured (AOR: 1.61, 95 % CI: 1.46—1.79, $p < 0.001$) are both significantly associated with increased odds of psychosocial health issues. Having no close friends significantly raises the odds of psychosocial health problems (AOR: 1.28, 95 % CI: 1.10—1.48, $p = 0.001$), compared to those with 3 or more friends. Parental factors such as never understanding the adolescent's problems (AOR: 1.44, 95 % CI: 1.27—1.63, $p < 0.001$) increased the risk of anxiety while and homework checking (AOR: 0.75 95 % CI: 0.67—0.84, $p < 0.001$) reduce the odds of psychosocial health issues. More than 4 h of sedentary behavior per day is associated with higher odds of psychosocial health problems (AOR: 1.50, 95 % CI: 1.32—1.71, $p < 0.001$). Suicidal ideation (AOR: 2.84, 95 % CI: 1.85—2.36, $p < 0.001$), plans (AOR: 1.36, 95 % CI: 1.21—1.54, $p < 0.001$), and attempts (AOR: 1.41, 95 % CI: 1.25—1.60, $p < 0.001$) are all significantly associated with increased odds of psychosocial health problems. Parental involvement is the protective factor.

The maps illustrate the prevalence of anxiety among adolescents across different regions of the world. The prevalence is categorized into five ranges, represented by different colors. Some countries in South America, Africa, and the Middle East show high levels of anxiety among adolescents. Countries like India, parts of Southeast Asia, and some regions in Africa and South America show moderate levels of anxiety.

Table 4 highlights the associations between various factors and adolescents' anxiety across different global regions. Older age (particularly 17 years) is significantly associated with higher risks of anxiety, especially in the Eastern Mediterranean, Southeast Asia, the Americas, and Western Pacific regions. Females generally exhibit a higher likelihood of experiencing these problems across all regions. Food insecurity is a notable factor, with adolescents who experience it more frequently showing a substantially higher likelihood of anxiety issues. Bullying, physical attacks, and serious injuries are consistently associated with elevated risks, while having close friends, parental involvement in

Table 2
Distribution of participants' characteristics by the level of anxiety based on pooled dataset.

Variables	Total	Never	Always	P-Value
Age in years				<0.001
11–12 years	16,089 (8.88)	14,657 (9.15)	1432(6.84)	
13 years	28,697 (15.84)	26,146 (16.32)	2551 (12.18)	
14 years	37,118 (20.48)	33,379 (20.83)	3739 (17.85)	
15 years	37,105 (20.48)	32,882 (20.52)	4223 (20.16)	
16 years	30,496 (16.83)	26,496 (16.53)	4000 (19.10)	
17 years	31,695 (17.49)	26,696 (16.66)	4999 (23.87)	
Gender of student				<0.001
Male	85,505 (47.43)	77,204 (48.40)	8301 (39.96)	
Female	94,765 (52.57)	82,295 (51.60)	12,470 (60.04)	
Being bullied				<0.001
No	117,292 (68.65)	107,451 (70.88)	9841 (51.11)	
Yes	53,557 (31.35)	44,143 (29.12)	9414 (48.89)	
Physically attacked				<0.001
No	119,654 (67.28)	108,183 (68.73)	11,471 (56.10)	
Yes	58,196 (32.72)	49,218 (31.27)	8978 (43.90)	
Physical fighting				
No	118,388 (66.34)	106,844 (67.65)	11,544 (56.26)	
Yes	60,078 (33.66)	51,102 (32.35)	8976 (43.74)	
Seriously injured				<0.001
No	90,502 (56.97)	83,376 (59.13)	7126 (39.92)	
Yes	68,363 (43.03)	57,639 (57,639)	10,724 (60.08)	
Peer supports				<0.001
Never	699(13.72)	585(13.36)	114(15.90)	
Sometimes or rarely	1808(35.49)	1527(34.88)	281(39.19)	
Most of the time or always	2588(50.79)	2266(51.76)	322(44.91)	
Parents check homework				<0.001
Never	45,158 (25.42)	38,642 (24.57)	6516 (31.91)	
Sometimes or rarely	65,798 (37.03)	58,754 (37.36)	7044 (34.49)	
Most of the time or always	66,720 (37.55)	59,858 (38.06)	6862 (33.60)	
Parents understand problem				<0.001
Never	46,642 (26.32)	39,703 (25.32)	6939 (34.02)	
Sometimes or rarely	67,698 (38.20)	60,219 (38.40)	7479 (36.67)	
Most of the time or always	62,860 (35.47)	56,883 (36.28)	5977 (29.31)	
Parent monitoring				<0.001
Never	39,505 (22.32)	33,804 (21.57)	5701 (28.08)	
Sometimes or rarely	66,385 (37.51)	58,732 (37.48)	7653 (37.69)	
Most of the time or always	71,099 (40.17)	64,147 (40.94)	6952 (34.24)	
Close Friends				<0.001
None	14,089 (7.84)	11,768 (7.39)	2321 (11.29)	
1–2 Friends	55,337 (30.79)	47,988 (30.15)	7349 (35.74)	
>3 Friends	110,277 (61.37)	99,387 (62.45)	10,890 (52.97)	
Sedentary behavior				<0.001
<1 h	59,031 (33.78)	53,119 (34.31)	5912 (29.63)	
1–2 h	50,435 (28.86)	45,627 (29.47)	4808 (24.10)	
3–4 h	30,484 (17.44)	26,895 (17.37)	3589 (17.99)	
>4 h	34,805 (19.92)	29,161 (18.84)	5644 (28.29)	
Adolescent obesity status				<0.001
Normal weight	123,854 (83.79)	110,451 (84.18)	13,403 (80.76)	
Overweight	15,504 (10.49)	13,479 (10.27)	2025 (12.20)	
Obesity	8451(5.72)	7282(5.55)	1169(7.04)	
Feeling lonely				<0.001
Never	157,174 (87.12)	144,773 (90.66)	12,401 (59.84)	
Always	23,236 (12.88)	14,915 (9.34)	8321 (40.16)	
Suicidal ideation				<0.001
No	146,148 (83.06)	133,599 (85.54)	12,549 (63.51)	
Yes	29,797 (16.94)	22,588 (14.46)	7209 (36.49)	
Suicidal plan				<0.001
No	149,069 (84.69)	135,633 (86.79)	13,436 (68.06)	
Yes	26,949 (15.31)	20,645 (13.21)	6304 (31.94)	
Suicidal attempt				<0.001
No	144,379 (81.21)	130,857 (83.07)	13,522 (66.74)	
Yes	33,409 (18.79)	26,670 (16.93)	6739 (33.26)	

checking homework, and monitoring appear to mitigate these risks in some regions. Sedentary behavior, especially more than four hours daily, is linked to increased anxiety problems globally. Obesity presents a more pronounced risk in the Western Pacific. Suicidal ideation, plans, and attempts are significant predictors of anxiety across all regions, with suicidal ideation showing the strongest association.

4. Discussion

This study represents the most comprehensive assessment of the prevalence and underlying causes of anxiety and food insecurity among school-going adolescents throughout 59 countries. Our results indicated that persistent food insecurity, female gender, and advanced age increased the likelihood of anxiety disorders among adolescents. Bullying, physical fighting, and serious injury caused adolescents to exhibit a heightened level of anxiety issues. Adolescents without close friends, whose parents did not check their homework and failed to understand their problems, were more prone to have anxiety disorders. Increased sedentary behaviors, suicidal ideation, planning, and attempts were all strongly linked with heightened probabilities of anxiety disorders among adolescents. Our findings, indicating a heightened likelihood of anxiety issues among older adolescents, correspond with findings from prior research conducted in Northern Spain (Casares et al., 2023). The idea that significant transitions from childhood to adulthood occur during late adolescence may contribute to this outcome. Adolescents experience substantial anxiety as they demand autonomy and make decisions about their future, including academic strategies, obtaining accommodations, developing intimate relationships, and starting work, all of which can lead to psychological health issues (Lipari

Table 3
Factors associated with adolescents' anxiety by logistic regression model by pooled data.

	Unadjusted Model		Adjusted Model	
Variables	UOR (95 % CI)	*P*-Value	AOR (95 % CI)	*P*-Value
Age in years				
11–12 years	RC		RC	
13 years	0.99 (0.87—1.14)	0.946	0.93 (0.75—1.15)	0.489
14 years	1.09 (0.95—1.24)	0.212	1.06 (0.87—1.29)	0.586
15 years	1.35 (1.1901.53)	<0.001	1.27 (1.05—1.53)	0.012
16 years	1.84 (1.62—2.09)	<0.001	1.77 (1.45—2.16)	<0.001
17 years	2.49 (2.21—2.80)	<0.001	2.45 (2.05—2.94)	<0.001
Gender of student				
Male	RC		RC	
Female	1.20 (1.13—1.28)	<0.001	1.51 (1.38—1.66)	<0.001
Food insecurity				
None	RC		RC	
Sometimes or rarely	1.43 (1.34—1.53)	<0.001	1.20 (1.09—1.32)	<0.001
Most of the time or always	3.33 (3.04—3.65)	<0.001	2.22 (1.94—2.55)	<0.001
Being bullied				
No	RC		RC	
Yes	2.81 (2.64—3.00)	<0.001	1.68 (1.52—1.85)	<0.001
Physically attacked				
No	RC		RC	
Yes	1.64 (1.54—1.74)	<0.001	1.01 (0.91—1.12)	0.844
Physical fighting				
No	RC		RC	
Yes	1.96 (1.84—2.08)	<0.001	1.26 (1.46—1.79)	<0.001
Seriously injured				
No	RC		RC	
Yes	2.51 (2.35—2.68)	<0.001	1.61 (1.46—1.79)	<0.001
Peer supports				
Never	0.91 (0.56—1.48)	0.714	Not Adjusted	
Sometimes or rarely	0.91 (0.61—1.36)	0.649		
Most of the time or always	RC			
Had close friends				
None	2.18 (1.97—2.40)	<0.001	1.28 (1.10—1.48)	0.001
1—2 Friends	1.47(1.38 (1.57)	<0.001	1.10 (0.99—1.21)	0.060
≥3 Friends	RC		RC	
Parents check homework				
Never	1.60 (1.49—1.73)	<0.001	1.06 (0.93—1.19)	0.390
Sometimes or rarely	1.03 (0.96—1.11)	0.409	0.75 (0.67—0.84)	<0.001
Most of the time or always	RC		RC	
Parents understand problem				
Never	1.75 (1.62—1.89)	<0.001	1.44 (1.27—1.63)	<0.001
Sometimes or rarely	1.17 (1.08—1.26)	<0.001	1.01 (0.90—1.13)	0.917
Most of the time or always	RC		RC	
Parent monitoring				
Never	1.51 (1.39—1.63)	<0.001	1.06 (0.93—1.20)	0.390
Sometimes or rarely	1.24 (1.15—1.34)	<0.001	0.99 (0.88—1.11)	0.889
Most of the time or always	RC		RC	
Sedentary behavior				
<1 h	RC		RC	
1–2 h	0.92 (0.85—0.99)	0.038	0.98 (0.87—1.09)	0.684
3–4 h	1.26 (1.16—1.38)	<0.001	1.16 (1.02—1.32)	0.026
>4 h	1.92 (1.77—2.08)	<0.001	1.50 (1.32—1.71)	<0.001
Adolescent obesity status				
Normal weight	RC		RC	
Overweight	1.25 (1.12—1.40)	<0.001	1.10 (0.94—1.28)	0.943
Obesity	1.27 (1.08—1.50)	0.005	1.08 (0.88—1.34)	0.483
Suicidal ideation				
No	RC		RC	
Yes	3.89(3.64 (4.14)	<0.001	2.84 (1.85—2.36)	<0.001
Suicidal plan				
No	RC		RC	
Yes	3.43 (3.19—3.68)	<0.001	1.36 (1.21—1.54)	<0.001
Suicidal attempt				
No	RC		RC	
Yes	3.37 (3.16—3.60)	<0.001	1.41 (1.25—1.60)	<0.001

and Hedden, 2014).

In terms of gender of the student, female adolescents experience a higher incidence of anxiety than their male counterparts, consistent with a study conducted in Iran (Mohammadi et al., 2020). Female adolescents may encounter anxiety because of gender based violence and discrimination inside educational institutions and society (Kim and Kim, 2024). Additionally, the elevated prevalence of mood disorders in females compared to males can be attributed to various biopsychosocial factors, including gender inequity, a propensity for females to internalize distress more than males, increased exposure to maltreatment, especially childhood sexual abuse and biological influences such as estrogen (Biswas et al., 2020).

Food inadequacy is an established sign of heightened susceptibility to mental health issues, particularly during adolescence and early adulthood (Carries et al., 2024). Our study demonstrated that anxiety intensified among adolescents experiencing food insecurity. Food insecurity negatively affects mental health and elevates the prevalence of anxiety symptoms. A systematic review and meta-analysis also revealed a significant correlation between food insecurity and heightened anxiety risk (Arenas et al., 2019). Given the inherent concerns about the food supply, food insecurity logically triggers anxiety, suggesting a correlation between adolescents' perceived anxiety and living in households experiencing food insecurity (Maynard et al., 2019). A significant number of younger individuals experience feelings of discomfort concerning the circumstances of food in their houses. Their experiences of powerlessness, despair, and guilt concerning the household food situation may have directly intensified their symptoms of anxiety. Preserving such a scenario may undermine the self-esteem and happiness of adolescents (Carries et al., 2024). While this study identifies a significant association between food insecurity and self-reported anxiety symptoms among adolescents, it is important to interpret these findings within the broader context of emotional development and environmental stress. Anxiety, like sadness or worry, is a normal human emotion and often serves as an adaptive response to challenging circumstances such as

Table 4
Factors associated with adolescents' anxiety by regions.

	Africa		Eastern Mediterranean		Southeast Asia		The Americas		Western Pacific	
Variables	AOR (95 % CI)	*P*-Value	AOR (95 % CI)	*P*-Value	AOR (95 % CI)	*P*-Value	AOR (95 % CI)	*P*-Value	AOR (95 % CI)	*P*-Value
Age in years										
11–12 years	RC		RC							
13 years	0.76 (0.43—1.34)	0.350	0.78 (0.59—1.02)	0.07	1.03 (0.70—1.50)	0.900	1.08 (0.66—1.78)	0.750	0.99 (0.68—1.45)	0.960
14 years	0.92 (0.55—1.55)	0.760	0.95 (0.74—1.23)	0.071	1.11 (0.77—1.59)	0.580	1.40 (0.87—2.25)	0.170	0.96 (0.67—1.37)	0.830
15 years	1.08 (0.66—1.78)	0.760	1.25 (0.97—1.62)	0.080	1.08 (0.76—1.54)	0.670	1.63 (1.01—2.63)	0.040	1.42 (1.00—2.02)	0.050
16 years	1.3 (0.80—2.11)	0.300	1.44 (1.12—1.86)	0.001	1.57 (1.01—2.43)	0.040	2.06 (1.27—3.34)	0.000	1.44 (0.99—2.07)	0.050
17 years	1.7 (1.06—2.71)	0.030	1.8 (1.39—2.33)	<0.001	1.85 (1.26—2.70)	<0.001	3.25 (1.88—5.61)	0.000	1.84 (1.29—2.64)	<0.001
Sex of student										
Male	RC		RC		RC		RC		RC	
Female	1.03 (0.89—1.2)	0.660	2.31 (2.02—2.64)	<0.001	1.37 (1.11—1.70)	<0.001	1.85 (1.52—2.23)	<0.001	1.28 (1.06—1.55)	0.010
Food insecurity										
None	RC		RC		RC		RC		RC	
Sometimes or rarely	1.63 (1.38—1.93)	<0.001	1.19 (1.03—1.37)	0.020	1.38 (1.10—1.74)	0.010	1.16 (0.97—1.38)	0.110	1.41 (1.16—1.71)	<0.001
Most of the time or always	2.61 (2.11—3.22)	<0.001	1.96 (1.61—2.40)	<0.001	2.35 (1.66—3.33)	<0.001	2.77 (1.96—3.90)	<0.001	2.49 (1.94—3.20)	<0.001
Being Bullied										
No	RC		RC		RC		RC		RC	
Yes	1.39 (1.2—1.62)	<0.001	1.57 (1.37—1.80)	<0.001	1.77 (1.39—2.24)	<0.001	1.82 (1.52—2.19)	<0.001	1.63 (1.34—1.98)	<0.001
Physically attacked										
No	RC		RC		RC		RC		RC	
Yes	1.13 (0.96—1.34)	0.150	1.12 (0.95—1.32)	0.160	1.27 (1.02—1.60)	0.040	1.26 (1.03—1.55)	0.030	1.29 (1.06—1.58)	0.010
Physical fighting										
No	RC		RC		RC		RC		RC	
Yes	0.96 (0.8—1.15)	0.650	1.24 (1.06—1.44)	0.001	1.27 (1.00—1.61)	0.050	1.19 (0.97—1.45)	0.100	1.07 (0.88—1.29)	0.510
Seriously injured										
No	RC		RC		RC		RC		RC	
Yes	1.53 (1.31—1.8)	<0.001	1.71 (1.49—1.96)	<0.001	1.7 (1.32—2.17)	<0.001	1.64 (1.37—1.95)	<0.001	1.67 (1.39—2.00)	<0.001
Had close friends										
None	1.01 (0.8—1.28)	0.930	1.73 (1.38—2.16)	<0.001	0.83 (0.55—1.25)	0.370	1.24 (0.91—1.69)	0.170	1.00 (0.73—1.38)	0.980
1—2 Friends	1.03 (0.88—1.21)	0.700	1.13 (0.99—1.30)	0.070	0.82 (0.63—1.07)	0.140	1.01 (0.84—1.22)	0.910	1.03 (0.85—1.24)	0.760
≥3 Friends	RC		RC		RC		RC		RC	
Parents check homework										
Never	0.80 (0.65—0.99)	0.040	1.01 (0.85—1.20)	0.950	1.51 (1.14—1.99)	<0.001	1.14 (0.90—1.43)	0.280	1.01 (0.78—1.30)	0.960
Sometimes or rarely	0.79 (0.66—0.96)	0.020	0.74 (0.63—0.88)	<0.001	0.79 (0.61—1.02)	0.070	0.95 (0.76—1.20)	0.680	0.82 (0.66—1.01)	0.060
Most of the time or always	RC		RC		RC		RC		RC	
Parents understand problem										
Never	1.12 (0.89—1.40)	0.320	0.99 (0.82—1.18)	0.890	1.59 (1.19—2.13)	<0.001	1.57 (1.22—2.02)	<0.001	1.12 (0.86—1.45)	0.420
Sometimes or rarely	0.92 (0.77—1.11)	0.390	0.88 (0.74—1.05)	0.170	1.09 (0.85—1.39)	0.500	0.86 (0.69—1.09)	0.210	0.89 (0.71—1.13)	0.350
Most of the time or always	RC		RC		RC		RC		RC	
Parent monitoring										
Never	0.92 (0.73—1.15)	0.460	1.09 (0.91—1.30)	0.340	0.85 (0.63—1.15)	0.290	1.07 (0.84—1.36)	0.580	1.24 (0.95—1.61)	0.110
Sometimes or rarely	0.79 (0.65—0.95)	0.020	0.97 (0.81—1.15)	0.690	1.08 (0.84—1.39)	0.530	0.96 (0.79—1.18)	0.710	1.13 (0.92—1.40)	0.250
Most of the time or always	RC		RC		RC		RC		RC	
Sedentary behavior										
<1 h	RC		RC		RC		RC		RC	
1–2 h	1.04 (0.87—1.23)	0.670	1.02 (0.87—1.2)	0.820	0.85 (0.65—1.11)	0.240	0.90 (0.71—1.14)	0.380	1.01 (0.80—1.28)	0.920
3–4 h	0.80 (0.63—1.02)	0.080	1.22 (1.00—1.47)	0.050	1.16 (0.87—1.55)	0.320	1.09 (0.85—1.39)	0.500	1.29 (1.01—1.66)	0.050

(*continued on next page*)

Table 4 (*continued*)

	Africa		Eastern Mediterranean		Southeast Asia		The Americas		Western Pacific	
Variables	AOR (95 % CI)	*P*-Value	AOR (95 % CI)	*P*-Value	AOR (95 % CI)	*P*-Value	AOR (95 % CI)	*P*-Value	AOR (95 % CI)	*P*-Value
>4 h	1.09 (0.88—1.36)	0.420	1.35 (1.12—1.62)	<0.001	1.72 (1.25—2.37)	<0.001	1.44 (1.14—1.81)	<0.001	1.58 (1.24—2.01)	<0.001
Adolescent obesity status										
Normal weight	RC		RC		RC		RC		RC	
Overweight	1.23 (0.93—1.63)	0.140	1.04 (0.84—1.30)	0.720	0.85 (0.55—1.34)	0.490	1.04 (0.79—1.36)	0.780	1.04 (0.82—1.33)	0.730
Obesity	0.94 (0.53—1.68)	0.840	1.16 (0.86—1.56)	0.340	0.79 (0.41—1.54)	0.500	1.04 (0.75—1.43)	0.820	2.40 (1.53—3.76)	<0.001
Suicidal ideation										
No	RC		RC		RC		RC		RC	
Yes	1.72(1.4—2.1)	<0.001	1.89 (1.58—2.26)	<0.001	2.33 (1.72—3.16)	<0.001	2.30 (1.8—2.93)	<0.001	1.98 (1.58—2.48)	<0.001
Suicidal plan										
No	RC		RC		RC		RC		RC	
Yes	1.52 (1.26—1.84)	<0.001	1.06 (0.88—1.28)	0.510	1.28 (0.91—1.8)	0.150	1.39 (1.1—1.77)	0.010	1.62 (1.29—2.04)	<0.001
Suicidal attempt										
No	RC		RC		RC		RC		RC	
Yes	0.9 (0.74—1.09)	0.270	1.50 (1.28—1.77)	<0.001	1.85 (1.31—2.62)	0.000	1.31 (1.04—1.64)	0.020	1.60 (1.27—2.02)	<0.001

hunger or uncertainty. The survey instrument used captures subjective emotional distress rather than clinical diagnoses, and thus the presence of anxiety symptoms does not necessarily indicate a mental disorder. Interventions aimed at reducing food insecurity may alleviate emotional distress, but should be framed within a holistic understanding of adolescent well-being, avoiding the conflation of normative emotional responses with psychopathology.

Our study identified bullying as a statistically significant risk factor for anxiety disorders in adolescents, which found that adolescents who experienced bullying exhibited higher anxiety levels, consistent with findings from other research conducted in Bangladesh (Khan and Khan, 2020). Bullying constitutes a substantial life stressor and adversely affects adolescents' psychosocial health problems, resulting in diminished self-esteem, severe depression, and other issues that may persist into adulthood (Juvonen and Graham, 2014; Rahman, 2025). Furthermore, childhood bullying victimization can impact three significant domains in the stage of adulthood: psychopathology, criminal behavior, and suicidal tendencies (Klomek et al., 2015).

The results of our study showed that adolescents who engaged in physical fighting and were seriously injured were more prone to anxiety than those who did not face any physical abuse. This tendency reflects findings from research in Finland demonstrating that adolescents who experienced physical abuse exhibited heightened levels of anxiety disorders (Rehan et al., 2017). Furthermore, these results complement metanalytical data indicating serious experiences of physical abuse were linked to an increased chance of acquiring anxiety disorders in adulthood (Norman et al., 2012). Traumatic experiences may adversely affect self-esteem, leading to feelings of depression, loneliness, and anxiety and these adolescents viewed suicide as their sole means of escaping further victimization (Swee et al., 2020).

The absence of close friends markedly increased the likelihood of anxiety disorders, in contrast to individuals with three or more friends. A study by Biswas et al. (2020) found a link between an increased risk of anxiety and the absence of close friends. A study in Guyana revealed that adolescents with close friends and compassionate parents exhibited a reduced likelihood of contemplating suicide. This indicates that parental and friend support mitigates psychosocial health problems by reducing the acceptability of mortality, reducing anxiety, and enhancing general mental well-being (Rudatsikira et al., 2007). Our findings revealed a significant association between parent engagement and adolescents' anxiety, indicating that adolescents whose parents sometimes or rarely check their homework have greater levels of anxiety. Parental involvement in their adolescent child's behavior is crucial for preventing high-risk activities and ensuring academic success, thus serving as a preventive measure for adolescents' psychosocial well-being (Ying et al., 2015). As found in this study, adolescents whose parents never understand their problems reported a higher likelihood anxiety disorder, a finding that was also reported in a study conducted in Vietnam (Nguyen et al., 2019). The causes include insufficient support from family members and parental aversiveness which is highly correlated with anxiety. Furthermore, inadequate parental support diminishes the likelihood of adolescents acquiring coping skills and the ability to manage adversities. These elevate the likelihood of developing anxiety disorders in adolescents (Khan and Khan, 2020). Parental understanding and having close friends serve as protective factors against anxiety.

The sedentary behavior of adolescents significantly influences anxiety disorders. Previous research indicated a strong correlation between sedentary behavior and anxiety among college students in China (Jiang et al., 2020). A systematic review and meta-analysis revealed that increased sedentary behavior is associated with depression, anxiety, and other internalizing illnesses in adolescents (Zhang et al., 2022). Prolonged sedentary behavior may hinder adolescents' social relationships as such behavior adversely affects mental health. Consequently, elevated hours of sedentary behavior, including excessive screen time, may result in social isolation and psychosocial health problems (Jiang et al., 2020).

Adolescents' suicidal ideation, plans, and attempts are strongly correlated with elevated probabilities of anxiety. Adolescents with anxiety disorders exhibited a greater propensity for suicidal ideation, plans, and attempts compared to their counterparts. These results align with prior research undertaken by Fang et al. (2024) and Khan and Khan (2020). Moreover, a direct correlation exists between anxiety and impulsivity: those experiencing anxiety are more prone to impulsive decision-making behaviors (Xia et al., 2017). Impulsivity enhances adolescents' involvement in violent and high-risk behaviors, including suicidal acts (McHugh et al., 2019). The researcher strongly advocates for an evidence-based primary therapy for psychosocial conditions such as anxiety and depression, specifically aimed at adolescents who are at risk of mitigating suicidal behaviors (Khan and Khan, 2020).

Both the strengths and weaknesses of this study have been recognized. The study's strengths include that it provides a thorough and wide-ranging review of the association between food insecurity and anxiety in teenagers by using data from 175,100 adolescents in 59 countries. This large sample size improves the findings' applicability and generalization to different populations and geographical areas. The study highlights both common and region-specific risk factors while providing insights into psychological health across a range of cultural, socioeconomic, and geographic contexts by encompassing nations from five WHO regions. Additionally, the study makes use of data from the Global School-based Student Health Survey (GSHS), a dependable and popular instrument for evaluating the health of adolescents, which raises the validity and consistency of the results.

Finally, in addition to highlighting food insecurity as a substantial risk factor, the study also finds other important characteristics that can guide focused interventions, including gender, age, peer victimization, and a lack of parental and peer support. The cross-sectional structure of the study makes it difficult to evaluate long-term consequences because it only collects data at one moment in time, which restricts the capacity to prove causation between food insecurity and anxiety. Because replies on anxiety and food insecurity may be influenced by social desirability, memory recall problems, or cultural standards, using self-reported data from adolescents may introduce bias. Although the study encompasses a wide range of nations, it does not go into great detail on the socioeconomic circumstances unique to each nation, which could have an impact on the incidence of anxiety and food insecurity. This limits the contextual interpretations of the findings. The results may have been impacted by participants' sociocultural backgrounds as a whole due to the sensitive nature of questions regarding mental health problems and their willingness to answer specific things (McKinnon et al., 2016). Countries could use GSHS forms that had been turned into their languages this means that the results may have been changed by translations into local languages (McKinnon et al., 2016), especially in instances when local languages lack equivalent terminology for the original concepts pertaining to psychological health issues. Some teenagers may have had trouble understanding the questionnaire because of their poor reading skills. In the same way, parental approval for their kids to take part might have depended on their capacity to read and sign the consent forms. Furthermore, despite the study's sample size, it might not fully capture the experiences of the most marginalized or vulnerable groups, such as those living in extreme poverty or not attending school, who may be more susceptible to mental health problems and food insecurity.

The strengths and limitations of this current study are properly endorsed. Firstly, the strengths of this present study worked with a large sample size of 179,937 adolescents across 59 countries, applying comprehensive and suitable statistical investigation. In addition, a large sample size augments the generalizability of the findings across various regions and populations. The study provides insights into anxiety across various cultural, socioeconomic, and geographic contexts by encompassing countries from five WHO regions, highlighting risk factors that are common and unique to each region. Additionally, the study makes use of data from the Global School-based Student Health Survey (GSHS), a dependable and popular database for evaluating the health of adolescents, which raises the validity and consistency of the results. Finally, by identifying a number of potential risk variables, including age, gender, food insecurity, peer victimization, and a lack of peer and parental support, the study can help relevant stakeholders create appropriate interventions. However, the cross-sectional structure of the study limits the capacity to establish a causal relationship between anxiety and food insecurity because it only records data at one particular moment in time, making it challenging to ascertain long-term impacts. Because replies on anxiety and food insecurity may be influenced by social desirability, memory recall problems, or cultural standards, using self-reported data from teenagers may introduce bias. Contextual interpretations of the data are limited because, although the study covers a wide range of countries, it does not go thoroughly into the socioeconomic conditions that are unique to each nation and may have an impact on the prevalence of food insecurity and psychological health disorders. Given the delicate nature of the questions about mental health issues and the participants' willingness to provide certain information, the results might have been influenced by their sociocultural backgrounds overall (McKinnon et al., 2016). Countries could use GSHS forms that had been turned into their languages this means that translations may have changed the results into local languages (McKinnon et al., 2016), especially in instances when local languages lack equivalent terminology for the original concepts pertaining to psychological health issues. Some adolescents may have had trouble understanding the questionnaire because of their poor reading skills. In a similar vein, parental approval for their kids to take part might have depended on their capacity to read and sign the consent forms. Furthermore, despite the study's sample size, it might not fully capture the experiences of the most marginalized or vulnerable groups, such as those living in extreme poverty or not attending school, who may be more susceptible to mental health problems and food insecurity. Additionally, this study employed an additive logistic regression model to examine the associations between psychosocial and behavioral factors and adolescent anxiety. While this approach offers interpretability and aligns with the exploratory nature of the analysis, it may not fully capture the complexity of relationships among variables. Specifically, potential interaction effects (e.g., between sex and bullying) and non-linear associations (e.g., age or sedentary behavior) were not modeled. These choices were made to maintain parsimony and avoid overfitting, given the large number of predictors. However, we acknowledge that excluding such terms may limit the model's ability to detect nuanced patterns. Future research should consider incorporating interaction and non-linear components to better understand the conditional and dynamic nature of these relationships.

Furthermore, an important limitation of this study is the potential influence of cultural factors on the expression and reporting of anxiety symptoms across the countries included in the sample. Although standardized survey instruments were used to promote consistency, cultural norms regarding emotional disclosure, mental health stigma, and linguistic interpretation may have affected how adolescents understood and responded to anxiety-related items. These cultural variations could introduce measurement bias and complicate cross-national comparisons. Future research should consider culturally adapted tools and mixed-method approaches to more accurately capture the lived experience of anxiety in diverse settings. Another notable limitation of this study is the absence of data on depressive symptoms, which often co-occur with anxiety and share overlapping features in adolescent populations. While the analysis focused on self-reported anxiety symptoms, the lack of a validated measure of depression restricts our ability to examine comorbidity or to differentiate between distinct emotional states. This may limit the interpretive depth of our findings, particularly in understanding the broader mental health implications of food insecurity. Future studies should incorporate comprehensive assessments of both anxiety and depression to more accurately reflect the complexity of adolescent emotional health.

5. Conclusion

The findings indicate a substantial regional variation in anxiety prevalence among adolescents, with the highest rates observed in the Eastern Mediterranean region (14.61 %), followed by the African region (11.43 %), and the lowest in Southeast Asia (3.78 %). The observed differences suggest that cultural, social, and environmental factors across these regions may influence anxiety rates. These results demonstrate considerable regional variations, indicating that adolescents in particular areas may be more susceptible to anxiety. Adolescent anxiety is influenced by several sociodemographic and psychological factors, as further evidenced by logistic regression analysis. Significant risk factors include being older, female, food insecure, being bullied, not having close friends, having parents who don't understand, being sedentary, and having suicidal thoughts. In order to reduce anxiety among adolescents, especially in areas where prevalence is highest, these findings highlight the necessity of customized interventions that address these characteristics.

CRediT authorship contribution statement

Md. Amirul Islam: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Tanjirul Islam:** Writing – review & editing, Writing – original draft. **Bristi Rani Saha:** Writing – review & editing, Writing – original draft. **Sakib Al Hassan:** Writing – review & editing, Writing – original draft. **Noman Hasan:** Writing – review & editing, Writing – original draft. **Md. Ashfikur Rahman:** Writing – original draft, Visualization, Validation, Supervision, Software, Methodology, Formal analysis, Data curation, Conceptualization.

IRB and informed consent statement

IRB and informed consent statements were not applicable because all data utilized in this study are de-identified secondary data collected from the school-Based Student Health Survey (GSHS) platform. There were no interactions or interventions with humans or animals in the current study.

Ethics statement

The GSHS received ethical approval from the regulatory agencies of the participating country, including the Ministry of Education and other relevant institutional ethics review committees. Participation necessitates parental approval, either in writing or verbal form. The survey was administered anonymously.

Declaration of Generative AI and AI-assisted technologies in the writing process

The author declares none, but as a non-native English speaker, Grammarly Pro was used to check the entire paper in order to remove outstanding grammatical errors and we ensured all edits furhter checked for clarity by all authors.

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Declaration of competing interest

The authors declare that they have no competing interests.

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Data availability

This study used publicly available Global School-Based Student Health Survey (GSHS) datasets which can be freely obtained from their respective websites. As a third-party user, we don't have permission to share the data publicly on any platform.

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